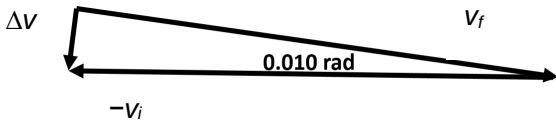


2(a)(i)		[1] correct diagram
2(a)(ii)	$\sin\left(\frac{0.010 \text{ rad}}{2}\right) = \frac{\Delta v / 2}{v} = \frac{\Delta v}{2v}$ $\Delta v = 2v \sin(0.0050 \text{ rad})$ $2v(0.0050) = 0.010v$	[1] [1] sine rule, cosine rule, arc length are acceptable
2(a)(iii)	The acceleration of P is pointing towards the centre of the circle.	[1]
2(a)(iv)	$t = \frac{s}{v} = \frac{r\theta}{v}$ $t = \frac{0.01r}{v}$	[1] [1]
2(a)(v)	$a = \frac{\text{change in velocity}}{\text{time taken}} = \frac{0.010v}{0.010(r/v)}$ $a = \frac{v^2}{r}$	[1]
2(b)(i)	As the bob spins round the pole, the radius r of the circular path will gradually decrease. <u>The centripetal force F_c on the bob and, hence, the tension T in the string will then increase.</u> $F_c = T = \frac{mv^2}{r}$ <u>T increases because m and v remain constant. The tension may eventually be large enough to cause the thread to snap.</u>	[1] [1]
2(b)(ii)	It is not possible due to the downwards gravitational force on P, the thread has to be slanted to allow for a vertical component of the tension to balance the weight.	[1]
2(a)	The straight line graph passes through the origin	